



KTU NOTES

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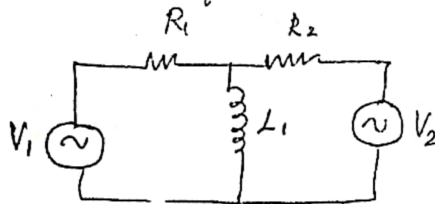
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Module 3

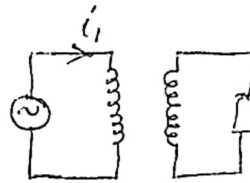
Transformed Circuits (s-domain)

Coupled Circuits

An electrical network containing two or more meshes is an example of coupled circuit. In this case the current flowing



Conductively coupled



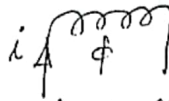
Inductively coupled

in one mesh produces a voltage drop across the common circuit element or elements between the meshes. This voltage drop in turn affects the current in the second mesh. Such circuits are called conductively coupled circuits.

Inductively coupled circuits: In this case an alternating current flowing in one circuit produces a magnetic flux and induces a voltage in the other circuit. As a result a current flows in the second circuit and a transfer of energy takes place from the 1st circuit to the second circuit.

Transformer coupled circuits are common examples of such circuits.

Self inductance, L



If the current passing through a circuit is changing, the flux linking the circuit also changes and hence according to Faraday's laws of Electromagnetic Induction, an emf is induced in the circuit. This emf is proportional to the time rate of change of current.

$$\text{ie } e \propto \frac{di}{dt}$$

$$= L \cdot \frac{di}{dt}, \text{ where the constant of proportionality}$$

' L ' is the self inductance of the circuit and 'e' is the self induced emf.

This self induced emf is also proportional to the rate of change of flux linkage.

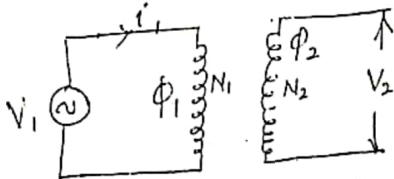
If the coil has N turns, the flux linkage is $N\Phi$ and

$$e = \frac{d(N\Phi)}{dt} = N \cdot \frac{d\Phi}{dt}$$

$$\therefore L \cdot \frac{di}{dt} = N \cdot \frac{d\Phi}{dt}$$

$$\text{or } L = N \cdot \frac{d\Phi}{di} = \frac{N \cdot \Phi}{I} \text{ Henry.}$$

Mutual Inductance, ' M '



Consider the circuit, the time varying current i_1 produces a flux Φ_1 .

If a second coil is placed near the

1st coil, some of the flux say Φ_{12} links coil 2.

Since the flux is changing at every instant, an emf will be induced in coil 2. ie $V_2 = N_2 \cdot \frac{d\Phi_{12}}{dt}$

Since Φ_{12} is produced by i_1 , V_2 will be proportional to

4. time rate of change of current i_1 , ie $V_2 \propto \frac{di_1}{dt}$

$$V_2 = M \frac{di_1}{dt}, \text{ where } M \text{ is the constant of proportionality}$$

M is called Mutual inductance and has the unit of Henry.

$$\therefore M \frac{di_1}{dt} = N_2 \cdot \frac{d\phi_{12}}{dt}$$

$$M = N_2 \cdot \frac{d\phi_{12}}{di_1} = \frac{N_2 \cdot \phi_{12}}{I_1}$$

Mutual Inductance is that property of the coil to produce an emf in a nearby coil by induction, when the current in the first coil changes.

Similarly if a current i_2 in the second coil produces a mutually induced emf in the first coil, then $V_1 = M \cdot \frac{di_2}{dt}$ &

$$V_1 = N_1 \frac{d\phi_{21}}{di_2} \quad \& \quad M = \frac{N_1 \phi_{21}}{I_2}$$

Coefficient of coupling

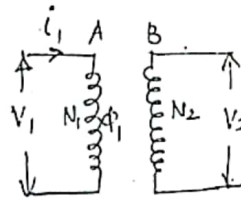
The fraction of the total flux produced by one coil that links with a second coil is called coeff. of coupling k .

$$\text{i.e. } k = \frac{\phi_{12}}{\phi_1} = \frac{\phi_{21}}{\phi_2}$$

Consider two magnetically coupled coils

Their self inductances are

$$L_1 = \frac{N_1 \phi_1}{I_1} \quad \& \quad L_2 = \frac{N_2 \phi_2}{I_2}$$



If a current i_1 produces a flux ϕ_1 in A and a portion ϕ_{12} links with the coil B.

$$\text{Then } M = \frac{N_2 \phi_{12}}{I_1} \quad \dots \dots \dots (1)$$

If a current i_2 in B produces a flux ϕ_2 in B and ϕ_{21} links with the coil A. Then

$$M = \frac{N_1 \phi_{21}}{I_2} \quad \dots \dots \dots (2)$$

Multiplying ① & ②, we get-

$$M^2 = \frac{N_1 \phi_{21}}{I_2} \cdot \frac{N_2 \phi_{12}}{I_1}, \text{ but } \phi_{12} = K \phi_1 \text{ \& } \phi_{21} = K \phi_2$$

$$\therefore M^2 = \frac{N_1 K \phi_2}{I_2} \cdot \frac{N_2 \cdot K \phi_1}{I_1}$$

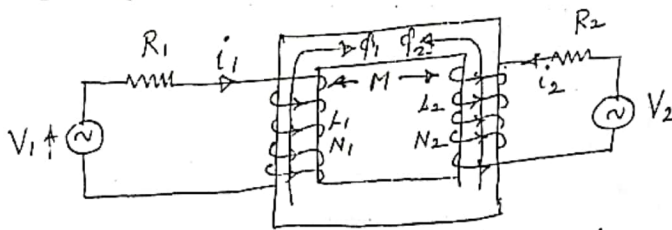
$$= K^2 \cdot \left(\frac{N_1 \phi_1}{I_1} \right) \left(\frac{N_2 \phi_2}{I_2} \right) = K^2 \cdot L_1 L_2$$

$$\therefore M = K \sqrt{L_1 L_2}$$

$K = \frac{M}{\sqrt{L_1 L_2}}$, where 'K' is the coefficient of coupling.

If the flux produced by one coil completely links with the other, then the value of 'K' is unity. Then the coils are said to be tightly coupled. If the flux of one coil does not link with the other, then $K=0$ and then the coils are magnetically isolated from each other.

Analysis of coupled circuits



Consider the coupled circuit shown above. The two coils are having a common core. If the fluxes ϕ_1 & ϕ_2 produced by i_1 & i_2 (for the assumed current directions) aid one another, then the signs of mutually induced voltages are the same as the signs of the voltages of self inductances.

If the fluxes oppose each other, sign of the mutually induced emfs are opposite to the self induced emfs.

To determine the direction of flux, apply right hand rule.

Right hand thumb rule

If the fingers wrap around in the direction of the assumed current, the thumb points in the direction of the flux.

In this case sign of 'M' term is opposite to sign of 'L' term

If we write the equations for each coil,

$$R_1 i_1 + L_1 \frac{di_1}{dt} - M \frac{di_2}{dt} = V_1$$

$$R_2 i_2 + L_2 \frac{di_2}{dt} - M \frac{di_1}{dt} = V_2$$

Fig. shows that there is no electrical coupling between the 2 coils. But the equation shows that there is coupling between the two loops.

Thus the coupling is said to be magnetic coupling.

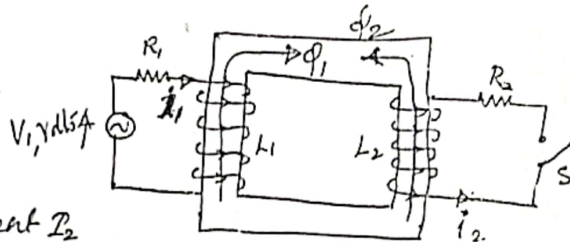
Natural current

Consider a coupled circuit having a passive secondary loop. Source V_1 drives a current i_1 , which produces a flux Φ_1 .

This current i_1 produces a mutually induced emf in the second coil. If the second coil circuit is closed, a current i_2 will flow in the circuit. According to Lenz's law, induced current always flows in such a direction as to oppose the cause which produces it.

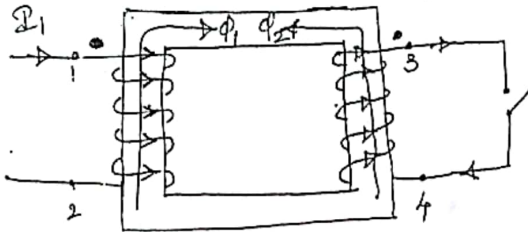
Here the cause producing the current i_2 is the flux in the 1st coil due to current i_1 .

$\therefore i_2$ flows in such a direction as to create a flux ~~in~~ opposing the main flux ~~opposing~~ due to i_1 . The mutually induced voltage $(M \frac{di_1}{dt})$ is the driving voltage for the second circuit. This voltage is present whether or not the circuit is closed.



Dot Rule for coupled circuits

The polarity of voltage of mutual induction is determined from the sketches of the core with winding senses and direction of current flows. To simplify the representation of coupled circuits, coils are marked with dots to identify the polarity of mutually induced emf.

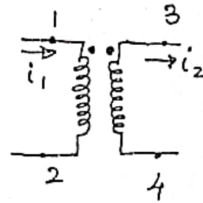


Selecting a current direction in one coil, a dot is placed at the terminal, where the current enters the winding. Applying the RH rule, the direction of flux is obtained. The flux of the other winding according to Lenz's law opposes the first flux. Then use the RH rule to determine the direction of natural current.

The voltage of mutual induction is positive at the terminal where the current leaves the winding. Hence a dot is placed at the terminal where the natural current leaves the winding.

Then the coupled coils can be represented as shown below.

This ~~second~~ terminal 3 is positive simultaneously with the terminal of the 1st coil where the initial current entered. With the instantaneous polarity



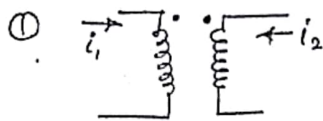
of the coupled coils given by the dots, the pictorial representation of the core with its winding sense is no longer needed.

The dot rule can be stated as follows

- ① when both assumed currents enter or leave a pair of dotted terminal, the sign of 'M' term is same as the sign of 'L' term.

If one current enters a dotted terminal and the other leaves a dotted terminal, the sign of 'M' term is opposite to the sign of 'L' term.

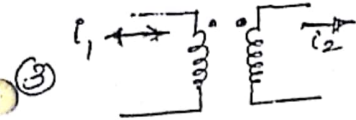
Dots indicate the terminals with the same instantaneous polarity.



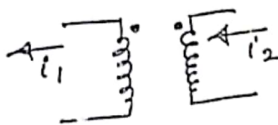
both the currents enters the dotted terminals
M & L terms have ~~opposite~~ ^{the same} sign



M & L have opposite signs



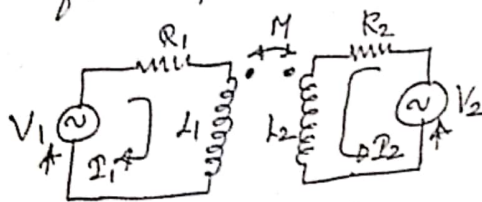
M & L have the same sign



M & L have opposite signs

Equivalent circuit of a coupled circuit

A conductively coupled equivalent circuit can be obtained for a coupled circuit.



Coupled circuit

writing voltage equations

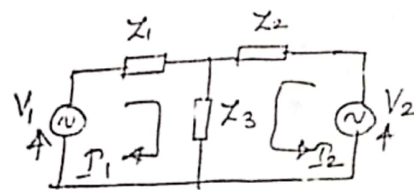
$$(R_1 + j\omega L_1)I_1 + j\omega M I_2 = V_1 \quad \text{--- (1)}$$

$$j\omega M I_1 + (R_2 + j\omega L_2)I_2 = V_2 \quad \text{--- (2)}$$

$$(R_1 + j\omega L_1)I_1 + j\omega M I_2 = V_1$$

$$j\omega M I_1 + (R_2 + j\omega L_2)I_2 = V_2$$

Comparing the coefficients of I_1 & I_2 in both the circuits, we get



Conductively coupled circuit

$$(Z_1 + Z_3)I_1 + Z_3 I_2 = V_1 \quad \text{--- (3)}$$

$$Z_3 I_1 + (Z_2 + Z_3)I_2 = V_2 \quad \text{--- (4)}$$

$$(R_1 + j\omega L_1) = Z_1 + Z_3 \quad \dots \quad (4)$$

$$(R_2 + j\omega L_2) = Z_2 + Z_3 \quad \dots \quad (5)$$

$$\& Z_3 = j\omega M \quad \dots \quad (6)$$

Substituting (6) in (4) & (5), we get

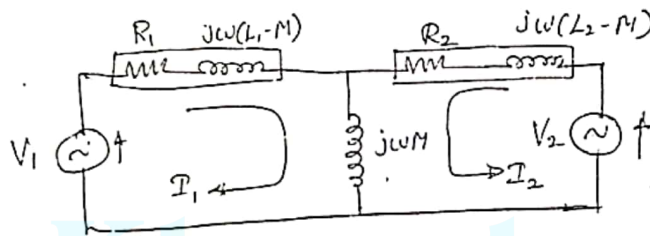
$$Z_1 + j\omega M = R_1 + j\omega L_1$$

$$\therefore Z_1 = R_1 + j\omega(L_1 - M)$$

$$\& Z_2 + j\omega M = R_2 + j\omega L_2$$

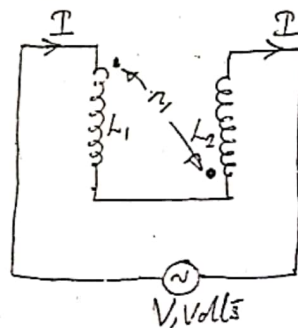
$$\therefore Z_2 = R_2 + j\omega(L_2 - M)$$

Then the conductively coupled equivalent circuit becomes

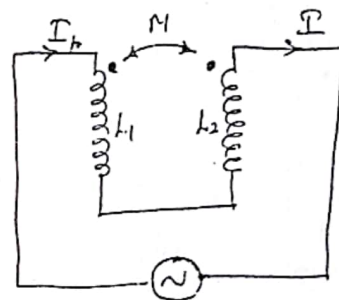


Note:
But a physically realizable equivalent circuit for any coupled circuit is not possible, especially when $M > L_1$ or L_2 which requires a negative value of inductance.

Coupled circuits in series



Case (a)



Case (b)

Case (a) - coils are connected with additive polarity. i.e. in this case, the current I enters at both the dotted terminals.

$\therefore M \& L$ have the same sign.

$$j\omega L_1 I + j\omega L_2 I + j2\omega M I = V$$

$$I [j\omega (L_1 + L_2 + 2M)] = V$$

$$I \cdot j\omega \cdot L_{eq} = V$$

$$\therefore \underline{L_{eq} = L_1 + L_2 + 2M}$$

In case (b)

$$j\omega L_1 I + j\omega L_2 I - 2 \cdot j\omega M I = V$$

$$j\omega (L_1 + L_2 - 2M) I = V$$

$$j\omega L_{eq} \cdot I = V$$

$$\therefore \underline{L_{eq} = L_1 + L_2 - 2M}$$

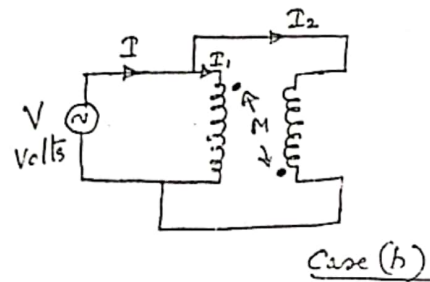
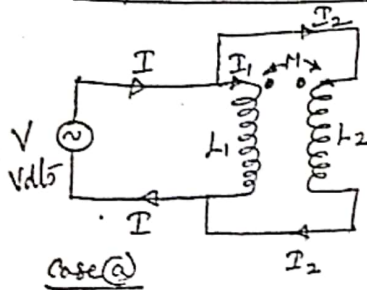
In general, $\boxed{L_{eq} = L_1 + L_2 \pm 2M}$

If fluxes aid each other, $L_{eq} = L_1 + L_2 + 2M$

If fluxes oppose each other, $L_{eq} = L_1 + L_2 - 2M$

If $M=0$, then $\boxed{L_{eq} = L_1 + L_2}$

Coupled circuits in parallel



Case (a) fluxes aid each other

writing voltage equations for each circuit

$$j\omega L_1 I_1 + j\omega M I_2 = V$$

$$j\omega M I_1 + j\omega L_2 I_2 = V$$

$$\begin{bmatrix} j\omega L_1 & j\omega M \\ j\omega M & j\omega L_2 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} V \\ V \end{bmatrix}$$

$$\begin{vmatrix} j\omega L_1 & j\omega M \\ j\omega M & j\omega L_2 \end{vmatrix} = -\omega^2 L_1 L_2 + \omega^2 M^2 = \omega^2 (M^2 - L_1 L_2)$$

$$\therefore P_1 = \frac{\begin{vmatrix} V & j\omega M \\ V & j\omega L_2 \end{vmatrix}}{\omega^2 (M^2 - L_1 L_2)} = \frac{V \cdot j\omega (L_2 - M)}{\omega^2 (M^2 - L_1 L_2)}$$

$$P_2 = \frac{\begin{vmatrix} j\omega L_1 & V \\ j\omega M & V \end{vmatrix}}{\omega^2 (M^2 - L_1 L_2)} = \frac{j\omega (L_1 - M) \cdot V}{\omega^2 (M^2 - L_1 L_2)}$$

$$P = P_1 + P_2 = j\omega \cdot V \cdot \frac{(L_1 + L_2 - 2M)}{\omega^2 (M^2 - L_1 L_2)}$$

$$\frac{V}{P} = \frac{\omega (M^2 - L_1 L_2)}{j \cdot (L_1 + L_2 - 2M)} = \frac{j\omega (L_1 L_2 - M^2)}{L_1 + L_2 - 2M}$$

$$= j\omega L_{eq}, \text{ where } L_{eq} = \frac{L_1 L_2 - M^2}{L_1 + L_2 - 2M}$$

$$\text{If the fluxes oppose, then } L_{eq} = \frac{L_1 L_2 - M^2}{L_1 + L_2 + 2M}$$

$$\text{In general } L_{eq} = \frac{L_1 L_2 - M^2}{L_1 + L_2 \pm 2M}. \text{ If } M=0, L_{eq} = \frac{L_1 L_2}{L_1 + L_2}$$

Ex- Two coils with a coeff. of coupling of 0.5 between them are connected in series so as to magnetise

a) in the same direction. b) in the opposite direction

The corresponding values of total inductances are for

a) 1.9 H and for (b) 0.7 H

Find the self inductance of the two coils and the mutual inductance between them.

Soln:

$$L_1 + L_2 + 2M = 1.9 \quad \text{--- ①}$$

$$L_1 + L_2 - 2M = 0.7 \quad \text{--- ②}$$

①-② gives

$$4M = 1.2$$

$$M = 0.3 \text{ H}$$



$$\text{Then } M/\sqrt{L_1 L_2} = K$$

$$K = 0.5$$

$$M = K\sqrt{L_1 L_2}, \therefore \sqrt{L_1 L_2} = M/K = \frac{0.3}{0.5} = 0.6$$

$$\therefore L_1 L_2 = 0.36$$

$$L_2 = 0.36/L_1$$

$$\text{Then } L_1 + \frac{0.36}{L_1} = 1.3$$

$$L_1^2 + 0.36 = 1.3 L_1$$

$$\text{or } L_1^2 - 1.3 L_1 + 0.36 = 0$$

$$L_1 = \frac{1.3 \pm \sqrt{(1.3)^2 - 4 \times 0.36}}{2} = \underline{\underline{0.9 \text{ H}}}$$

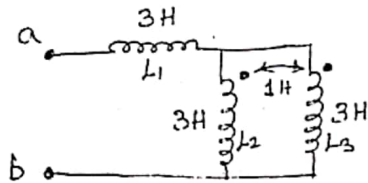
$$\& L_2 = \underline{\underline{0.4 \text{ H}}}$$

eg.2. Two coils of inductances 4 & 6 H are connected in parallel. If their mutual inductance is 3H, calculate the equivalent inductance of the combination if (1) M assists L & (2) M opposes L

$$1. L_a = \frac{L_1 L_2 - M^2}{L_1 + L_2 - 2M} = \underline{\underline{3.75 \text{ H}}}$$

$$2. L_b = \frac{L_1 L_2 - M^2}{L_1 + L_2 + 2M} = \underline{\underline{0.94 \text{ H}}}$$

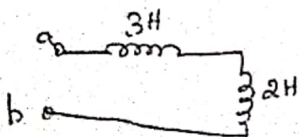
Eq.3. Find the equivalent inductance across the terminals 'ab'



$$L_{eq} = 3 + [3 \parallel 3] \text{ with } M=1\text{H}$$

$$L_{eq} \text{ of } 3\text{H} \parallel 3\text{H} = \frac{L_2 L_3 - M^2}{L_2 + L_3 - 2M} \quad [\text{fluxes aid}]$$

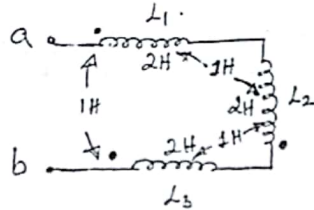
$$= \frac{3 \times 3 - 1}{3 + 3 - 2} = \frac{8}{4} = \underline{\underline{2 \text{ H}}}$$



$$\text{Now } L_{ab} = 3 + 2 = \underline{\underline{5 \text{ H}}}$$

$\therefore$ no mutual inductance between L_1 & L_2

eq. A.



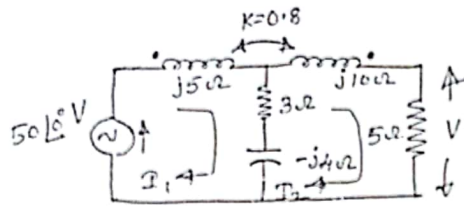
Find L_{eq}

$$L_{eq} = L_1 + L_2 + L_3 \pm 2M_{12} \pm 2M_{23} \pm 2M_{31}$$

$$= 2 + 2 + 2 - 2 + 2 - 2$$

$$= \underline{\underline{4H}}$$

Fig. 5
(a) Compute the voltage 'V' across 5Ω resistor for the coupled circuit shown below.



$$M = K \sqrt{L_1 L_2}$$

$$j\omega M = K \sqrt{j\omega L_1 \cdot j\omega L_2} = 0.8 \sqrt{j5 \times j10}$$

$$= \underline{\underline{j5.66\Omega}}$$

For loop 1, Loop 2

$$(3 + j5 - j4)I_1 - (3 - j4)I_2 - j5.66I_2 = 50\angle 0^\circ$$

$$\text{or } (3 + j1)I_1 - (3 + j1.66)I_2 = 50\angle 0^\circ \quad \text{--- (1)}$$

For loop 2

$$(8 + j6)I_2 - (3 + j1.66)I_1 = 0 \quad \text{--- (2)}$$

$$\begin{bmatrix} 3+j1 & -3-j1.66 \\ -3-j1.66 & 8+j6 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 50 \\ 0 \end{bmatrix}$$

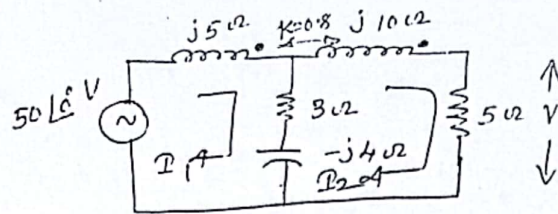
$$I_2 = \frac{\begin{vmatrix} 3+j1 & 50 \\ -3-j1.66 & 0 \end{vmatrix}}{\begin{vmatrix} 3+j1 & -3-j1.66 \\ -3-j1.66 & 8+j6 \end{vmatrix}} = \frac{50(3+j1.66)}{(3+j1)(8+j6) - (3+j1.66)^2}$$

$$= \frac{50(3+j1.66)}{(12.7156 + j35.96)}$$

$$= \underline{\underline{8.62 \angle -24.79^\circ A}}$$

$$V = I_2 \times 5 = \underline{\underline{43.1 \angle -24.79^\circ V}}$$

2) Calculate the voltage when the polarity position of dots is reversed.



$$(3+j1)I_1 - (3-j4)I_2 + j5.66 I_2 = 50$$

$$(3+j1)I_1 + (-3+j9.66)I_2 = 50 \quad \text{--- (1)}$$

$$-(8+j6)I_2 - (3-j4)I_1 + j5.66 I_1 = 0$$

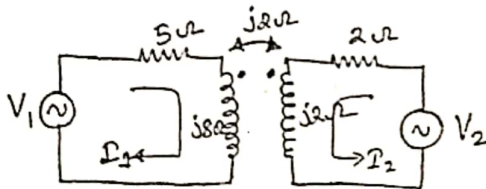
$$(-3+j9.66)I_1 + (8+j6)I_2 = 0 \quad \text{--- (2)}$$

$$\begin{bmatrix} 3+j1 & -3+j9.66 \\ -3+j9.66 & 8+j6 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 50 \\ 0 \end{bmatrix}$$

$$I_2 = \frac{\begin{vmatrix} 3+j1 & 50 \\ -3+j9.66 & 0 \end{vmatrix}}{\begin{vmatrix} 3+j1 & -3+j9.66 \\ -3+j9.66 & 8+j6 \end{vmatrix}} = \underline{\underline{3.82 \angle -112.12^\circ \text{ A}}}$$

$$V_2 = 5 I_2 = \underline{\underline{19.1 \angle -112.12^\circ \text{ V}}}$$

3) For the coupled circuit shown, find the ratio $\frac{V_2}{V_1}$, which results in zero current I_1 .



$$I_1(5+j8) + j2 I_2 = V_1 \quad \text{--- (1)}$$

$$(2+j2)I_2 + j2 I_1 = V_2 \quad \text{--- (2)}$$

when $I_1 = 0$, (1) & (2) becomes

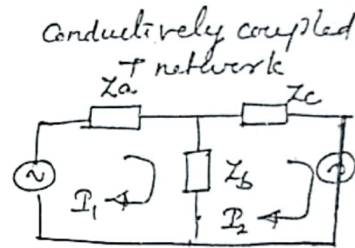
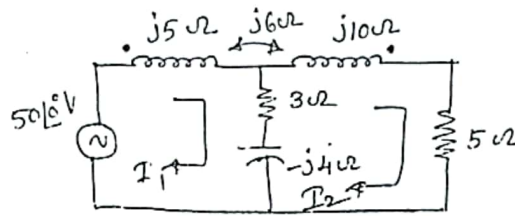
$$j2 I_2 = V_1$$

$$(2+j2)I_2 = V_2$$

$$\therefore \frac{V_2}{V_1} = \frac{2+j2}{j2} = \frac{-j2(2+j2)}{4} = \frac{4-j4}{4}$$

$$= \underline{\underline{1-j1}}$$

- ⑦ Obtain a conductively coupled equivalent circuit for the magnetically coupled circuit shown below.



For the inductively coupled case

$$(3+j1)I_1 - (3+j2)I_2 = 50 \quad \text{--- (1)}$$

$$-(3+j2)I_1 + (8+j6)I_2 = 0 \quad \text{--- (2)}$$

For the T-network

$$(Z_a + Z_b)I_1 - Z_b I_2 = 50 \quad \text{--- (3)}$$

$$-Z_b I_1 + (Z_c + Z_b)I_2 = 0 \quad \text{--- (4)}$$

Comparing the coefficients of I_1 & I_2

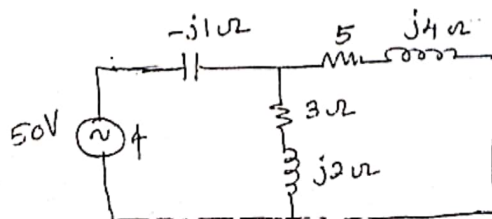
$$Z_a + Z_b = 3+j1$$

$$Z_b = (3+j2)$$

$$Z_a = (3+j1) - (3+j2) = \underline{\underline{-j1\Omega}}$$

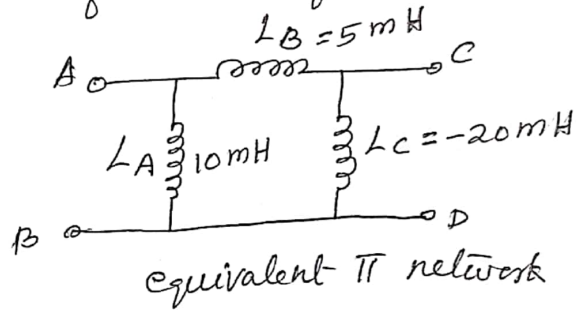
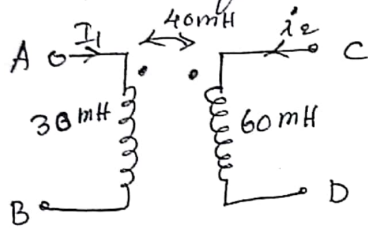
$$Z_b + Z_c = 8+j6$$

$$Z_c = 8+j6 - (3+j2) = \underline{\underline{(5+j4)\Omega}}$$



equivalent circuit

Find the equivalent π network of the transformer shown below

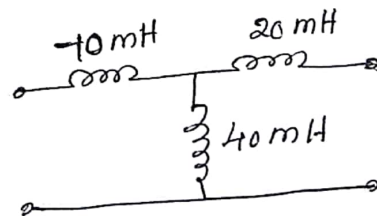
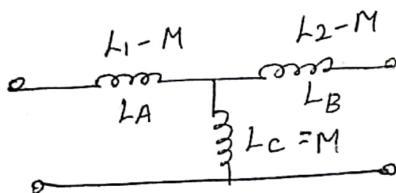


where
$$L_A = \frac{L_1 L_2 - M^2}{L_2 - M} = \underline{\underline{10 \text{ mH}}}$$

$$L_B = \frac{L_1 L_2 - M^2}{M} = \underline{\underline{5 \text{ mH}}}$$

$$L_C = \frac{L_1 L_2 - M^2}{L_1 - M} = \underline{\underline{-20 \text{ mH}}}$$

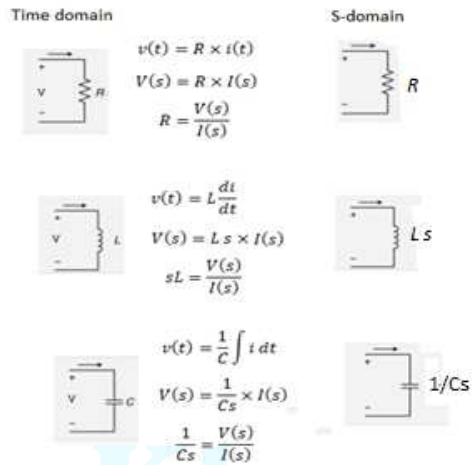
Also obtain T network



MODULE 3-TRANSFORMED CIRCUITS IN S-DOMAIN

Circuit Elements in the s Domain

With zero initial conditions



With initial conditions

A resistor in the s domain

By operational Laplace transform:

relation in the time domain

$$v(t) = R \cdot i(t).$$

$$L\{v(t)\} = L\{R \cdot i(t)\} = R \cdot L\{i(t)\},$$

$$\Rightarrow V(s) = R \cdot I(s).$$

$$R = \frac{V(s)}{I(s)}$$

An inductor in the s domain

- *iv*-relation in the time domain

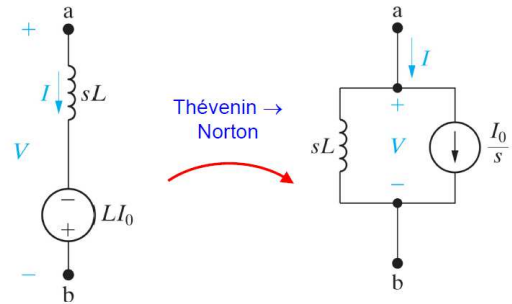
$$v(t) = L \cdot \frac{d}{dt} i(t).$$

- By operational Laplace transform:

$$\begin{aligned} L\{v(t)\} &= L\{L \cdot i'(t)\} = L \cdot L\{i'(t)\}, \\ \Rightarrow V(s) &= L \cdot [sI(s) - \underbrace{I_0}_{\text{initial current}}] = sL \cdot I(s) - LI_0 \end{aligned}$$

Equivalent circuit of an inductor

- Series equivalent:
- Parallel equivalent:



A capacitor in the s domain

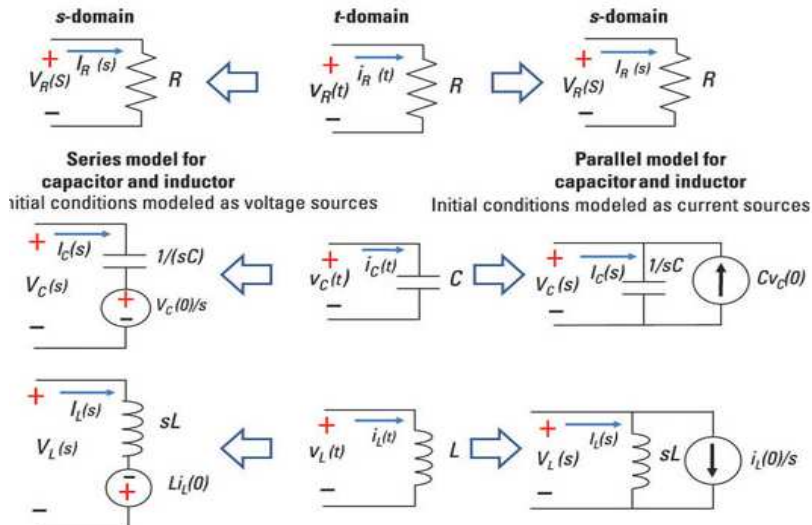
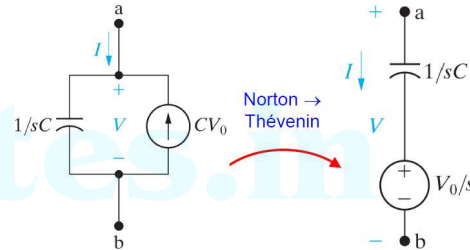
- *iv*-relation in the time domain

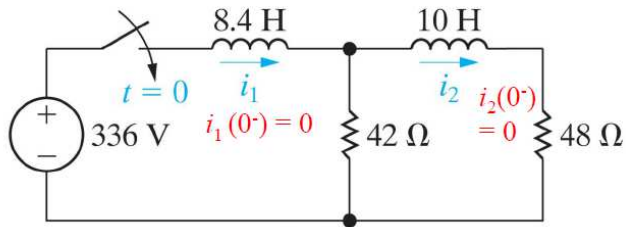
$$i(t) = C \cdot \frac{d}{dt} v(t).$$

- By operational Laplace transform:

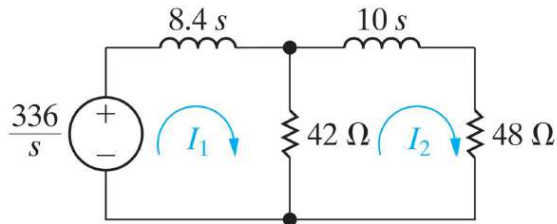
$$\begin{aligned} L\{i(t)\} &= L\{C \cdot v'(t)\} = C \cdot L\{v'(t)\}, \\ \Rightarrow I(s) &= C \cdot [sV(s) - \underbrace{V_0}_{\text{initial voltage}}] = sC \cdot V(s) - CV_0. \end{aligned}$$

- Parallel equivalent:
- Series equivalent:

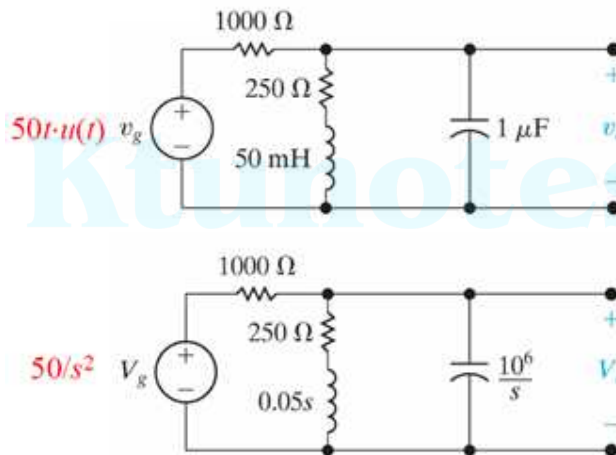




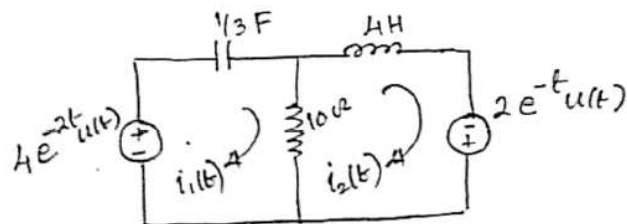
TIME DOMAIN

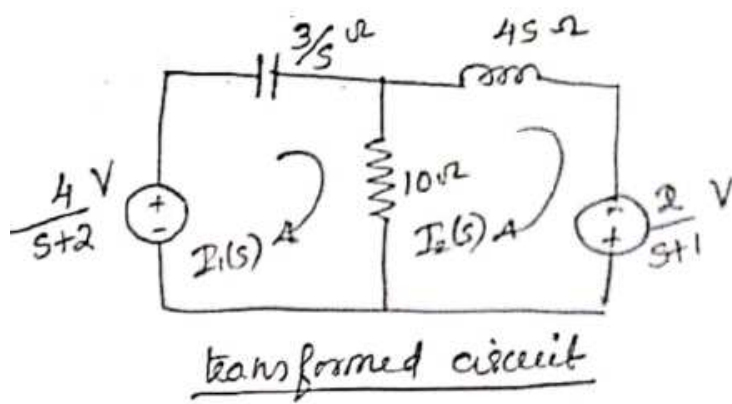


S-DOMAIN



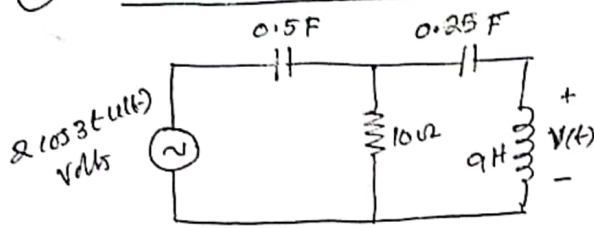
Draw the transformed circuit of the time domain circuit shown below



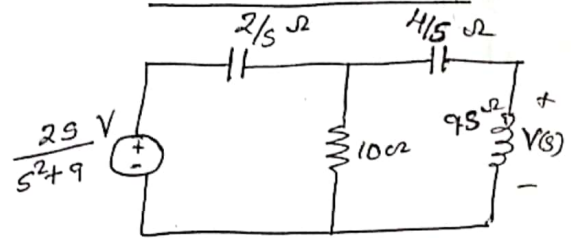


Ktunotes.in

② Time domain circuit

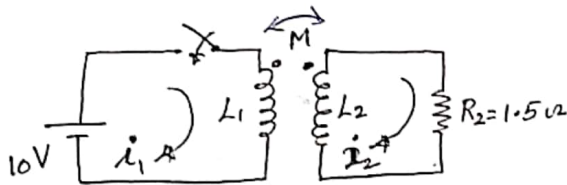


S-domain circuit

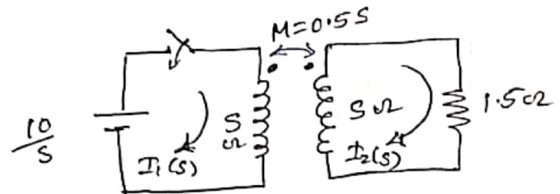


Examples - Soln in S-domain

- ① Write mesh equations in S-domain for the network, when a 10V source is switched on at $t=0$ sec. The primary and secondary self inductances are $L_1 = L_2 = 1$ H and $M = 0.5$ H



S-domain circuit



S-domain equations are
for primary

$$\frac{10}{s} - s I_1(s) + 0.5 s I_2(s) = 0$$

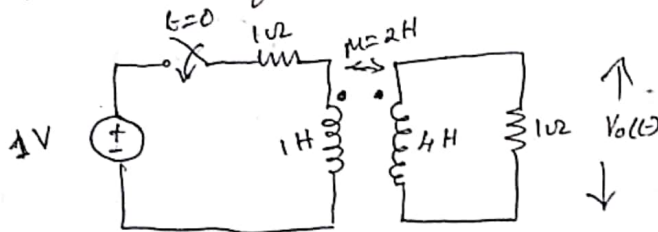
$$s I_1(s) - 0.5 s I_2(s) = \frac{10}{s} \quad \text{--- ①}$$

For secondary

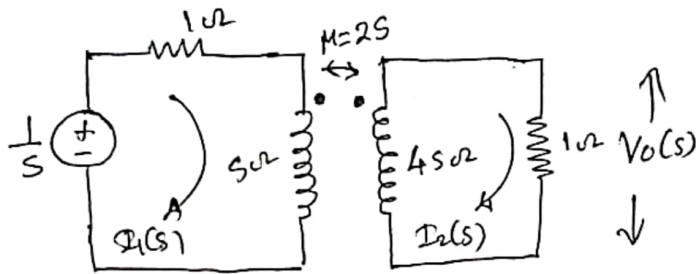
$$-1.5 I_2(s) - s I_2(s) + 0.5 s I_1(s) = 0$$

$$-0.5 s I_1(s) + (s + 1.5) I_2(s) = 0 \quad \text{--- ②}$$

- ② In the circuit shown in fig, the switch is closed at $t=0$ sec. Draw the transformed circuit and determine the voltage $V_o(t)$ for $t \geq 0$



Obtain the voltage across 1Ω , $V_o(t)$ for the S-domain circuit shown below



For the 1st loop, KVL eqn is

$$1/s - I_1(s) - 5sI_1(s) + 2sI_2(s) = 0$$

$$(s+1)I_1(s) - 2sI_2(s) = 1/s \quad \text{--- (1)}$$

$$-I_2(s) - 4sI_2(s) + 2sI_1(s) = 0$$

$$(4s+1)I_2(s) - 2sI_1(s) = 0 \quad \text{--- (2)}$$

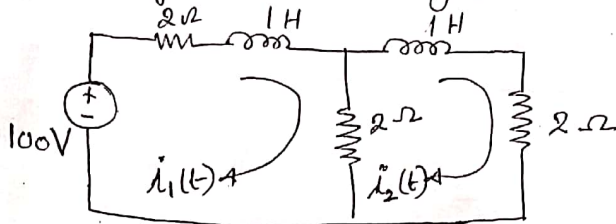
$$\begin{bmatrix} s+1 & -2s \\ -2s & 4s+1 \end{bmatrix} \begin{bmatrix} I_1(s) \\ I_2(s) \end{bmatrix} = \begin{bmatrix} 1/s \\ 0 \end{bmatrix}$$

$$I_2(s) = \frac{\begin{vmatrix} s+1 & 1/s \\ -2s & 0 \end{vmatrix}}{\begin{vmatrix} s+1 & -2s \\ -2s & 4s+1 \end{vmatrix}} = \frac{2}{(s+1)(4s+1) - 4s^2} = \frac{2}{4s^2 + s + 4s + 1 - 4s^2} = \frac{2}{5s+1} = \frac{2/5}{s+1/5}$$

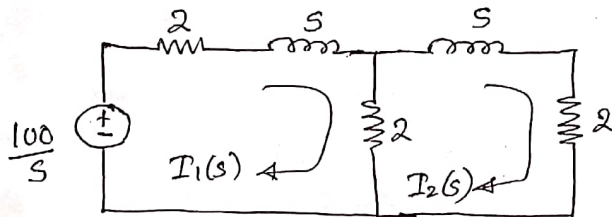
$$i(t) = \mathcal{L}^{-1} \left[\frac{2/5}{s+1/5} \right] = \frac{2}{5} e^{-t/5} \text{ A}$$

$$V_o(t) = 1\Omega \times \frac{2}{5} e^{-t/5} = \frac{2}{5} e^{-t/5} \text{ Volts}$$

1. In the circuit shown in figure, draw the transformed circuit and determine the current $i_2(t)$ using mesh analysis. Assume zero initial conditions. (KTU July 2017)



Soln: Transformed circuit is drawn as shown below.



$$\begin{bmatrix} 4+s & -2 \\ -2 & 4+s \end{bmatrix} \begin{bmatrix} I_1(s) \\ I_2(s) \end{bmatrix} = \begin{bmatrix} 100/s \\ 0 \end{bmatrix}$$

$$I_2(s) = \frac{\begin{vmatrix} 4+s & 100/s \\ -2 & 0 \end{vmatrix}}{\begin{vmatrix} 4+s & -2 \\ -2 & 4+s \end{vmatrix}} = \frac{200/s}{(4+s)^2 - 4}$$

$$= \frac{200}{s(s^2 + 8s + 12)}$$

$$= \frac{200}{s(s+2)(s+6)}$$

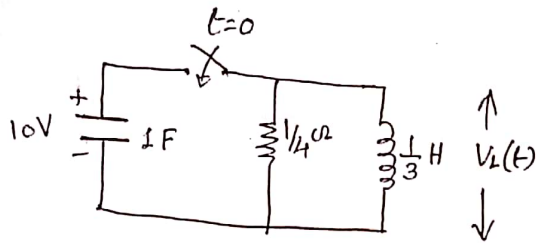
$$I_2(s) = \frac{200}{s(s^2 + 8s + 12)} = \frac{200}{s(s+2)(s+6)} = \frac{A}{s} + \frac{B}{s+2} + \frac{C}{s+6}$$

On solving, $A = \frac{50}{3}$, $B = -25$ & $C = \frac{25}{3}$

$$I_2(s) = \frac{50}{3} \cdot \frac{1}{s} - \frac{25}{s+2} + \frac{25}{3} \cdot \frac{1}{s+6}$$

$$i_2(t) = \left(\frac{50}{3} - 25e^{-2t} + \frac{25}{3}e^{-6t} \right) A$$

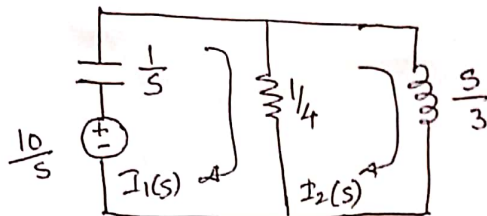
2. In the given circuit, Capacitor C has an initial voltage charge $V_0 = 10\text{ V}$ and at the same instant current in the inductor is zero. Switch is closed at time $t=0$. Obtain an expression for the voltage across the inductor L .
[KTU Jan 2017]



$$V(0) = 10\text{ V}$$

$$i_L(0) = 0\text{ A}$$

Soln: Transformed circuit is



$$\begin{bmatrix} \frac{1}{s} + \frac{1}{4} & -\frac{1}{4} \\ -\frac{1}{4} & \frac{s}{3} + \frac{1}{4} \end{bmatrix} \begin{bmatrix} I_1(s) \\ I_2(s) \end{bmatrix} = \begin{bmatrix} 10/s \\ 0 \end{bmatrix}$$

$$I_2(s) = \frac{\begin{vmatrix} \frac{1}{s} + \frac{1}{4} & 10/s \\ -1/4 & 0 \end{vmatrix}}{\begin{vmatrix} \frac{1}{s} + \frac{1}{4} & -1/4 \\ -1/4 & s/3 + \frac{1}{4} \end{vmatrix}} = \frac{\frac{10}{4s}}{\left(\frac{1}{s} + \frac{1}{4}\right)\left(\frac{s}{3} + \frac{1}{4}\right) - \frac{1}{16}}$$

$$= \frac{30}{(s+1)(s+3)}$$

$$V_L(s) = Ls \cdot I_2(s)$$

$$= \frac{s}{3} \cdot \frac{30}{(s+1)(s+3)} = \frac{10s}{(s+1)(s+3)} = \frac{A}{(s+1)} + \frac{B}{(s+3)}$$

$$A = -5 \text{ and } B = 15$$

$$= \frac{-5}{s+1} + \frac{15}{s+3}$$

$$\therefore V_L(s) = \frac{-5}{s+1} + \frac{15}{s+3} \quad \cdot \text{Inverse transform gives}$$

$$\therefore V_L(t) = (-5e^{-t} + 15e^{-3t}) \text{ Volts}$$

Complex frequency of Network Function

The complex variable $S = \sigma + j\omega$ is defined as the complex frequency. ' ω ', the imaginary part of ' S ' is interpreted as radian frequency and it appears in time domain equations in the form of $\sin \omega t$ or $\cos \omega t$. Radian frequency has the dimension of radians per sec and may be expressed in terms of frequency ' f ' in Hz or in terms of time period ' T ' in seconds by the equation $\omega = 2\pi f = \frac{2\pi}{T}$

The real part of complex frequency is reper frequency.

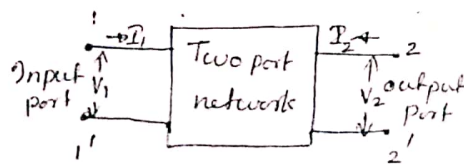
$\sigma = 0$ describes the steady state behaviour & $S = (\sigma + j\omega)$ describes the transient behaviour. The solutions of ~~differe~~ differential equations for the networks have given rise to time domain functions of the form $K e^{St}$, where $S = \sigma + j\omega$

The imaginary part ' ω ' corresponds to oscillations.

The real part ' σ ' corresponds to exponential decay or increase

Network function

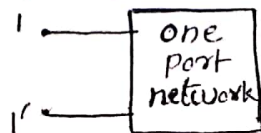
A network can be represented by a rectangle.



A large no. of physical networks like transformers are having two ports i.e. two terminal pairs (i.e. a port is a pair of terminals)

A port is defined as any pair of terminals into which energy is supplied or from which energy is ~~taken~~ withdrawn or where the network variables are measured.

A network having only one pair of terminals is called a one-port network.



In the two port network, the terminal pair 1-1' is  called input port, which is connected to the driving force or input source.

The terminal pair 2-2' is called output port is connected to a load.

Two ports containing no sources in their branches are called passive ports. eg - power transmission lines & transformers.

Two ports containing sources in their branches are called active ports.

A voltage and current are assigned to each of the two ports, V_1 & I_1 at the input port & V_2 & I_2 at the output port. It is also assumed that I_1 & I_2 are entering the networks at the upper terminals 1 & 2 respectively.

Thus the variables of a two port networks are

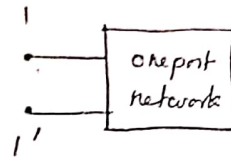
V_1, V_2 & I_1, I_2 i.e. a total of 4 quantities.

Thus in a network containing ~~a single~~ sources, the operator which relates current or voltage at different (or function) parts of the network is called a network function.

For a one-port network,

the ratio of Laplace transform of the applied voltage to the Laplace transform of the input current

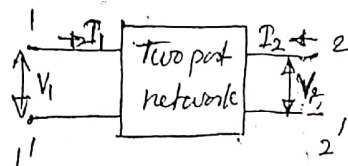
is ~~called~~ known as driving point impedance $Z(s)$



$$\text{i.e. } Z(s) = \frac{V(s)}{I(s)} \quad \& \quad Y(s) = \frac{I(s)}{V(s)} = \frac{1}{Z(s)}, \text{ where}$$

$Y(s)$ is called driving point admittance.

Network functions of two-port Network



A large no. of network functions are possible in this case.

Driving point impedance at port 1 is, $Z_{11}(s) = \frac{V_1(s)}{I_1(s)}$

Driving point admittance at port 1 is, $Y_{11}(s) = \frac{I_1(s)}{V_1(s)}$

Similarly driving point impedance at port 2 is $Z_{22}(s) = \frac{V_2(s)}{I_2(s)}$

and driving point admittance at port 2 is, $Y_{22}(s) = \frac{I_2(s)}{V_2(s)}$

There are four transfer functions possible for a 2-port network.

The ratio of Laplace transform of one quantity at the output port to the Laplace transform of another quantity at the input port is known as transfer function

(Assume zero initial conditions)

These transfer functions are

1. Voltage transfer ratio, $G_{21}(s) = \frac{V_2(s)}{V_1(s)}$

2. Current transfer ratio, $\alpha_{21}(s) = \frac{I_2(s)}{I_1(s)}$

3. Transfer impedance, $Z_{21}(s) = \frac{V_2(s)}{I_1(s)}$ ✓

4. Transfer admittance, $Y_{21}(s) = \frac{I_2(s)}{V_1(s)}$ ✓

Procedure to find Network functions for a two port network

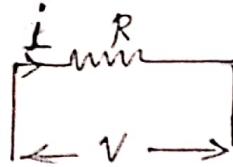
1. Write loop or node equations
2. Write Laplace transformed equations, assuming all initial conditions being set to zero.

③ Then network function is found by algebraically solving for the ratio of response ~~transform~~ transform to the input transform.

For 'R' network

$$Ri = V$$

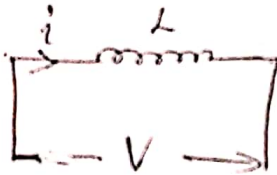
$$RI(s) = V(s)$$



network function

$$\boxed{\frac{V_i(s)}{I_i(s)} = R}$$

L-network

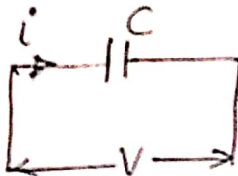


$$V = L \frac{di}{dt}$$

$$V(s) = L \cdot s I(s)$$

$$\therefore \boxed{\frac{V(s)}{I(s)} = L \cdot s}$$

C-network



$$V = \frac{1}{C} \int i dt$$

$$V(s) = \frac{1}{Cs} I(s)$$

$$\therefore \boxed{\frac{V(s)}{I(s)} = \frac{1}{Cs}}$$

Similarly for RL network

$$\boxed{\frac{V(s)}{I(s)} = R + Ls}$$

For RC network

$$\boxed{\frac{V(s)}{I(s)} = R + \frac{1}{Cs}}$$

And for RLC network

$$\boxed{\frac{V(s)}{I(s)} = R + Ls + \frac{1}{Cs}}$$

Poles and Zeros of Network function

In general, every network function is a quotient of polynomials in s . (i.e. ratio of polynomials in s)

i.e. Network function, $N(s) = \frac{P(s)}{Q(s)}$, which will have the general form as

$$N(s) = \frac{P(s)}{Q(s)} = \frac{a_0 s^n + a_1 s^{n-1} + a_2 s^{n-2} + \dots + a_{n-1} s + a_n}{b_0 s^m + b_1 s^{m-1} + b_2 s^{m-2} + \dots + b_{m-1} s + b_m}$$

where a 's & b 's coefficients are real and positive for networks having passive elements and no controlled sources.

The degree of numerator polynomial is ' n ' and that of denominator polynomial is ' m '.

$\therefore$ The equation $P(s)=0$, has ' n ' roots, say $z_1, z_2, \dots, z_n$

and $Q(s)=0$, has ' m ' roots, say $p_1, p_2, \dots, p_m$.

Then $N(s)$ may be written as $N(s) = \frac{a_0}{b_0} \frac{(s-z_1)(s-z_2) \dots (s-z_n)}{(s-p_1)(s-p_2) \dots (s-p_m)}$

$$\& N(s) = H \frac{(s-z_1)(s-z_2) \dots (s-z_n)}{(s-p_1)(s-p_2) \dots (s-p_m)}, \text{ where}$$

$z_1, z_2, \dots, z_n$ & $p_1, p_2, \dots, p_m$ are complex frequencies.

H is the scale factor and $H = \frac{a_0}{b_0}$

when the variable s has the values $z_1, z_2, \dots, z_n$

the network function vanishes and therefore the complex frequencies $z_1, z_2, \dots, z_n$ are known as the zeros of the network function.

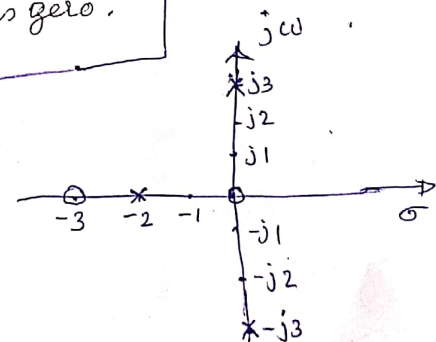
Thus zeros of a network function are defined as the values of ' s ' for which the function approaches zero.

~~we~~ locate poles and zeros of

$$Z(s) = \frac{4s(s+3)}{(s^2+9)(s+2)} \quad \text{in} \quad \begin{cases} \text{zeros} \\ s=0 \& \\ s=-3 \end{cases}$$

Complex s plane

$$\begin{cases} \text{poles at } s = \pm j3 \\ s = -2 \end{cases}$$



When the variable 's' has the values $p_1, p_2 \dots p_m$, the network function becomes infinite and therefore the complex frequencies $p_1, p_2 \dots p_m$ are called poles of the network function.

Thus poles of a network function are defined as the values of 's' for which the function approaches infinity.

Thus a network function is completely specified by its poles, zeros and scale factor.

Poles and zeros are very useful quantities in describing the behaviour of any network function.

The poles and zeros are critical frequencies, because at poles, the network function becomes infinite, while at zeros, the network function becomes zero. At every other frequency, the network function has a finite non-zero value.

When a pole or zero is ^{not} repeated, it is said to be simple or distinct.

When 'r' poles or zeros have the same value, the pole or zero is said to be of multiplicity 'r'.

The poles and zeros at $s = \infty$ are also assigned a degree or multiplicity.

If $n > m$, then $(n-m)$ zeros are at $s = \infty$. and for

$m > n$, then $(m-n)$ poles are at $s = \infty$

If the poles and zeros at infinity are taken into account in addition to the finite poles and zeros, the total no. of poles and zeros, ~~the total no. of poles and zeros~~ are equal.

